Cholesky decomposition

$$x_1 + 2x_2 + 3x_3 = 5$$
$$2x_1 + 8x_2 + 22x_3 = 6$$
$$3x_1 + 22x_2 + 82x_3 = -10$$

sblution:-

The given system of equations is of the form $Ax = b$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 8 & 22 \\ 3 & 22 & 82 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 5 \\ 6 \\ -10 \end{pmatrix}$$

✓ symmetric $\rightarrow A = A^T$

Now the matrix A is decomposed as

$$A = L L^T$$

$$A = u^T u$$

$$u_{ii} = \sqrt{a_{ii} - \sum_{k=1}^{i-1} u_{ki}^2}$$

$$u_{ij} = \frac{a_{ij} - \sum_{k=1}^{i-1} u_{ki} \cdot u_{kj}}{u_{ii}}$$

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$A = U^T U$

$Ax = b$

$U^T U x = b.$

upper $U^T x = d \Rightarrow$ Forward substitution.

$U d = b \Rightarrow$ Back substitution.

Lower

$$U_{11} = \sqrt{a_{11} - \sum_{k=1}^{0} u_{0k}^2}$$

$$U = \begin{vmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{vmatrix}$$

$$u_{11} = \sqrt{1} = 1$$

$$u_{12} = \frac{a_{12} - \sum_{k=1}^{0} u_{01} u_{02}}{u_{11}}$$

$$= \frac{2}{1} = 2$$

$$u_{13} = \frac{a_{13}}{u_{11}} = 3$$

$$u_{22} = \sqrt{a_{22} - \sum_{k=1}^{1} u_{ki}^2}$$

$$= \sqrt{a_{22} - u_{12}^2}$$

$$= \sqrt{8 - 2^2}$$

$$= \sqrt{8 - 4}$$

$$= \sqrt{4} = 2$$

$$u_{23} = \frac{a_{23} - \sum_{k=1}^{1} u_{ki} \cdot u_{kj}}{u_{22}}$$

$$u_{23} = \frac{a_{23} - (u_{12} \cdot u_{13})}{u_{22}}$$

$$= \frac{22 - (2 \times 3)}{2}$$

$$= \frac{22 - 6}{2}$$

$$= 11 - 3$$

$$= 8$$

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$$u_{33} = \sqrt{a_{33} - \sum_{k=1}^{2} u_{ki}^2}$$

$$= \sqrt{a_{33} - (u_{13}^2 + u_{23}^2)}$$

$$= \sqrt{82 - (3^2 + 8^2)}$$

$$= \sqrt{82 - (73)}$$

$$= \sqrt{9}$$

$$= 3$$

$$u = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 2 & 8 \\ 0 & 0 & 3 \end{pmatrix}, \qquad u^T = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 2 & 0 \\ 3 & 8 & 3 \end{pmatrix}$$

$$u^T d = b$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 2 & 2 & 0 \\ 3 & 8 & 3 \end{pmatrix} \begin{pmatrix} d_1 \\ d_2 \\ d_3 \end{pmatrix} = \begin{pmatrix} 5 \\ 6 \\ -10 \end{pmatrix}$$

$$d_1 = 5$$

$$d_2 = \frac{6 - 10}{2} \Rightarrow d_2 = -2$$

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$$3 \times 5 + 8 \times -2 + 3d_3 = -10$$

$$15 - 16 + 3d_3 = -10.$$

$$3d_3 = -9$$

$$d_3 = -3.$$

$$d = \begin{pmatrix} 5 \\ -2 \\ -3 \end{pmatrix}$$

$$Ux = d.$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 2 & 3 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 5 \\ -2 \\ -3 \end{pmatrix}$$

$$3x_3 = -3 \Rightarrow x_3 = -1$$

$$x_2 = \frac{-2 - (3 \times -1)}{2} = \frac{-2 + 3}{2} = 0.5$$

$$x_1 = 5 - (2 \times 0.5 + 3 \times -1).$$

$$= 5 - (-2).$$

$$= 7.$$

$$x = \begin{pmatrix} 7 \\ 0.5 \\ -1 \end{pmatrix}.$$

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